Summary in Graph

Exam Summary (GO Classes GATE CS | Theory of Computation | Context Free Grammar | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (33) out of 334 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 Theory of Computation

Consider the context-free grammar G :

$S \rightarrow aS \mid Aa$

$A \rightarrow bS|Aa|\lambda$

The number of 2 length strings generated by G is ?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Is there for all context-free grammars a right linear grammar that produces the same language?

A. Yes, you can find such a grammar by converting to an automaton, and then converting back to a grammar.
B. Yes, because all right linear grammars are context-free grammars.
C. No, because not all context-free languages are regular.
D. No, because not all regular languages are context-free.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Consider the grammar G:
$\mathrm{S} \rightarrow \mathrm{AB}$
$\mathrm{A} \rightarrow \mathrm{aA} \mid \lambda$
$\mathrm{B} \rightarrow \mathrm{bB} \mid \lambda$
Which of the following is/are correct?

A. G is regular.
B. G is not regular.
C. L(G) is regular.
D. $L(G)$ is not regular.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following statements are true?

A. If L is generated by a context-free grammar that is not a regular grammar, then L must not be a regular language.
B. If L is generated by a regular grammar then L must be a regular language.
C. Every context-free language is a nonregular language.
D. If a language L is accepted by a nondeterministic finite automaton, then there must be some context-free grammar that generates L .

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Suppose L is a language generated by a regular grammar, and consider the following statements:

(i) L is a regular language.
(ii) L is a context-free language.
(iii) There is a Turing machine that accepts L.
Which of the following is correct?

A. Only statement (i) is true.

B. Only statements (i) and (iii) are true.
C. All 3 statements are true.
D. None of the 3 statements is true.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Let $\Sigma = \{a, b\}$, and let $L = \{a^n w a^n : n \geq 1, w \in \Sigma^*\}$. Consider the following statements:
(i) L has regular expression $a^*(a+b)^*a^*$.
(ii) L is a non-regular language.

(iii) L has context-free grammar

$$S \rightarrow aSa|aS|bS \mid aa$$

where S, X are nonterminals.
(iv) L has context-free grammar

$$S \rightarrow aSa \mid aXa$$
$$X \rightarrow aX|bX|\Lambda$$

where S, X are nonterminals.

Which of the following is true?

A. Only statement (ii) is correct.
B. Only statements (ii) and (iv) are correct.
C. Only statement (iv) is correct.
D. None of the 4 statements is correct.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

The language $A = \{b^n a^n \mid n \geq 0\}$ satisfies which of the following?

A. A has regular expression b^*a^*.
B. A has regular expression $(ba)^*$.
C. A has CFG $G = (V, \Sigma, R, S)$, with $V = \{S\}, \Sigma = \{a, b\}, R = \{S \rightarrow bSa \mid \varepsilon\}$, and starting variable S
.
D. A is not context-free.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

The following Context-Free Grammar (CFG) :

$S \rightarrow aB|bA$

$A \rightarrow a|aS|bAA$

$B \rightarrow b|bS|aBB$

will generate

A. Odd numbers of $a's$ and odd numbers of $b's$
B. Even numbers of $a's$ and even numbers of $b's$
C. Equal numbers of $a's$ and $b's$
D. Different numbers of $a's$ and $b's$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

A context free grammar for $L = \{w \mid n_0(w) > n_1(w)\}$ is given by:

A. $S \rightarrow 0 \mid 0S \mid 1SS$
B. $S \rightarrow 0S \mid 1S \mid 0SS \mid 1SS \mid 0 \mid 1$
C. $S \rightarrow 0 \mid 0S \mid 1SS \mid S1S \mid SS1$
D. $S \rightarrow 0S \mid 1S \mid 0 \mid 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

A regular grammar for the language $L = \{a^n b^m \mid \text{ n is even and m is even }\}$ is given by

A. $S \rightarrow aSb \mid S_1; S_1 \rightarrow bS_1a \mid \lambda$
B. $S \rightarrow aaS \mid S_1; S_1 \rightarrow bSb \mid \lambda$
C. $S \rightarrow aSb \mid S_1; S_1 \rightarrow S_1ab \mid \lambda$
D. $S \rightarrow aaS \mid S_1; S_1 \rightarrow bbS_1 \mid \lambda$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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